Type MS · Improve reliability with repeats, means and larger samples

Answer 1 of 9 · Question 5 part c.ii · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

5 (a) (i)	A = eardrum; B = cochlea;		2
(ii)	Any two from the following • collects sound waves/vibrations/determines direction of sound; • directs them through the auditory / ear canal; • to the eardrum;		2
(b) (i)	• axes labels; • units; • bars correctly plotted; • scale;		4
(ii)	• (20 – 14) 6; • (6 ÷20) -0.30; • (x100) -30.00%	Allow ecf is minus symbol not given (Max 2 marks)	3
(c) (i)	Any two from the following • gender variation; • some people may have an ear defect;		2
(ii)	EITHER • check results; • with (data) from another group; OR		2

Type MS · Improve reliability with repeats, means and larger samples

Answer 1 of 9 · Question 5 part c.ii · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

	• repeat the experiment; • calculate mean;		
			Total 15

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 2 of 9 · Question 5 part a.iii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

5	(a)	(i)	175.9 + 178.5 + 166.3 + 167.6 ÷ 4;		1
		Clip with table	172.1;	Full marks for correct final answer Reject 172	1
		(ii)	Height increases with age (for males / females / both); Males are taller/females are shorter (at all ages); Females increase in height greater compared with males between the ages of 12 and 18; Greatest height increase between ages 12 to 14;		Max 2
		(iii)			1
			Use more students in each age group; Use a greater age range of students; Use ages with 1 year gap/13,15,17;	Allow other valid improvements Ignore repeats	1
	(b)	(i)	71 ÷ 1.46^2;		1
		Clip with table	33/33.3;	Full marks for final correct answer One mark for correct calculation with incorrect figures	1
		(ii)	student Y is obese;	ECF from b(i)	1
		CLIP (b)(i) with (b)(ii)	higher risk of diabetes/heart disease/arthritis/cancers;	Allow other medical condition linked to obesity	1

Total question 5 = 10 marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 2 of 9 · Question 5 part a.iii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 3 of 9 · Question 5 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
5 (a)	(i)	• axes labelled (1) • suitable scale (1) • independent variable on X axis (1) • correct plots (2) • suitable curve (1)	-1 for each incorrect plot	6
	(ii)	• mean/average calculated (1) • reliable / reproducible / repeatable (1) • eliminate anomalies (1)		2
(b)	(i)	70 x 100 (1) 90 78%/77.78/77.8 (1)	correct answer = full marks	2
	(ii)	• travels in the blood (1) • time to reach/affect nerve (1)	R heart	2
(c)		• carries impulses to heart muscle (1) • reduces rate of contraction (1)		2

Total for question = 14 marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 3 of 9 · Question 5 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 4 of 9 · Question 7 part b.iii · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

Number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 4 of 9 · Question 7 part b.iii · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 5 of 9 · Question 1 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
1	(a) (i)	<table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table> • two column table; • headings; • correct order of results (high to low/ORA);		3 marks
	(ii)	• no repeats/only done once;		1 mark
	(iii)	• results based on judgement/subjective/qualitative;		1 mark
	(iv)	• repeats; • measure volume of oxygen; • using gas syringe/other suitable apparatus;		3 marks
	(b) (i)	untreated/room temperature;		1 mark
	(ii)	control;		1 mark
	(iii)	• reduced/low enzyme activity; • not at optimum pH; • active site changed; • fewer enzyme substrate complexes/substrate won't fit;		4 marks

Total 14 marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 5 of 9 · Question 1 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 6 of 9 · Question 2 part b.iii · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

2 (a)	(i)	in the following order: • proteases; • protein Q; • amino acids;	Reject protein P	3
	(ii)	stomach/ small intestine/ pancreas/ duodenum/ ileum;		1
	(iii)	DNA / mRNA / messenger RNA;		1
(b)	(i)	• as pH increases activity increases; • to pH2/peaks at pH2/to optimum; • decreases from pH2;	Allow increases and then decreases for one mark	3 exp
	(ii)	Any two from: • enzyme is denatured (at pH2); • change in shape of active sites; • no enzyme-substrate complexes formed/no binding with substrate;		2 exp
	(iii)	• repeat; • calculate a mean/average/remove anomalous result		2 grad
	(iv)	Any one from: • temperature; • substrate concentration/volume; • enzyme concentration/volume;		1 grad

Total for Question 2 = 13 Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 7 of 9 · Question 4 part b.i · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Improve reliability with repeats, means and larger samples

Answer 7 of 9 · Question 4 part b.i · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 8 of 9 · Question 2 part e · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part e in this mark scheme extract

Number					
2	(a)	(i)	• axes labels; • plots; • scale;	axes correct way round use aty least half of the graph paper for the plotted results ignore extrapolation of line	3
		(ii)	line of best fit;		1
		(iii)	As time increases pH decreases;	ECF from an incorrectly drawn graph	1
	(b)		Any three from: • lipase; • digests/breaks down fat (in milk); • to fatty acids; • which lowers pH/increases acidity;		3
	(c)		time;	allow mins/minutes	1
	(d)		Any two from: • temperature; • volume/concentration of enzyme/substrate (solution)/volume of milk; • type of milk;	ignore amount	2
	(e)		repeat the experiment		1

Total Marks 12

Type MS · Improve reliability with repeats, means and larger samples

Answer 8 of 9 · Question 2 part e · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 9 of 9 · Question 7 part d · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

number				
7 (a)	(i)	• temperature of incubation; • volume of milk; • composition of nutrients in agar/type of agar;		3
	(ii)	size of bacterial growth;		1
(b)	(i)	$\frac{7.1}{5} = 1.42$; $\frac{36.5}{5} = 7.3$;	One mark for each correct answer	2
	(ii)	• more bacteria in B; • at start of investigation; • because extra added to tube; • some bacteria in A/milk which grew;	Allow more growth in B/growth increases more quickly in B Allow reference to contamination e.g.from air	4
(c)		• transfer loop sterilised/heated in flame; • transfer performed quickly;	Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment	2
(d)		• repeat investigation; • using other organisms;		2

Total 14 marks

Type MS · Improve reliability with repeats, means and larger samples

Answer 9 of 9 · Question 7 part d · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks